ESA Full content:

Ariane 3 Rocket:

Ariane 3 was a European expendable carrier rocket. This rocket was used for 11 launches between 1984 and 1989.This rocket belongs to the Ariane family and was derived from the Ariane 2 rocket. The Ariane 3 followed the same basic design as the Airane 1 but has some modifications. This rocket has a two solid-fuelled PAP strap-on-booster which was used to augment the first stage on life-off. The core of this rocket is based on its earlier version ariane 2 and is just made from the upgrades for ariane 2.The first stage was powered by four Viking 2B bipropellant engines, this used a dinitrogen teroxide oxideser. The second stage was powered by the Viking 4B which used the same fuel-oxidiser. The third stage was powered by the cryogenic fuelled HM7B engine. On some of the rockets, additional fourth stage was added and was powered by the Mage 2 engine. This rocket has a height of 49.13m with a diameter of 3.8m.This rocket has a mass of 234,000kg.The payload to GTO is 2,700kg.

Ariane 4 Rocket:

The Ariane 4 is an expendable launch system. This is a first flight on 15th June 1988 until the final flight, which was performed on 15th February 2003.This rocket has attained 113 successful launches out of 116 conducted. In 1982, the Ariane 4 programme was approved by the European Space Agency (ESA).The Ariane 4 was the ultimate development from the preceding members of the Ariane family. Compared with the Ariane 2 and Ariane 3, the Ariane 4 featured stretched first and third stages. The rocket was used in a number of variants - it could be fitted with two or four additional solid and a liquid fuelled booster rockets. The engine used in this rocket is Viking 6.this rocket can use either of 0,2 and 4 no. of boosters. This rocket has a height of 58.72m with a mass of 3.8m.This rocket has a mass of 240,000kg.The payload to LEO of this rocket is 5,000kg and 2,000kg to GTO orbit.

S. no	Payload to GTO	Launches	Success	Failures
1.	2100kg	7	7	0
2.	2930kg	15	15	0
3.	3480kg	13	13	1
4.	4720kg	40	39	1

Ariane 5 Rocket:

Ariane 5 is a European heavy-lift launch vehicle that is part of the Ariane rocket family. This is also an expendable launch system designed by the French government space agency. It is used to deliver payloads into GTO orbit or LEO orbit. The Ariane project was instrumental in helping to overcome the European space crisis. Ariane 5 rockets are manufactured under the authority of ESA and the French space agency. Ariane 5 succeeded Ariane 4 but was not derived form it. Unlike all the ariane rockets, this rocket was a new model without any modifications of the other Arianes. Ariane 5's biogenic H173 main stage is called the EPC. it consists of a large tank 30.5m high with two compartments, one for liquid oxygen and the other for the liquid hydrogen. This rocket has a height of 46-52m with a diameter of 5.4m.This rocket has a mass of 777,000kg and has 2 stages. The payload to LEO is around 16,000kg and payload to GTO is 6,950kg.

S. no	Date	Payload	Launch outcome
1.	4th June 1996	Cluster	Failure
2.	30th October 1997	MaqSat-H,TEAMSAT,MaqSat-B,YES	Partial failure
3.	21st October1998	MaqSAt-3,ARD	Success
4.	10th December 1999	XMM-Newton	Success
5.	21st March 2000	INSAT-3B,AsiaStar	Success